

PAPER

GTVelo: Cell Fate Prediction with Graph Attention

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Abstract

Untangling the development and differentiation process in single-cell data remains a fundamental challenge in single-cell biology. We introduce GTVelo, a graph-based deep learning framework for temporal cell state prediction using velocity-weighted Graph Transformers. GTVelo constructs cell graphs with edges weighted by both transcriptomic similarity and RNA velocity alignment, enabling the model to learn directional developmental relationships. A Graph Transformer aggregates information across cells, capturing long-range dependencies that single-cell features alone cannot reveal. We evaluate GTVelo on developmental single-cell datasets; embryonic hematopoiesis, neurogenesis, and erythroid differentiation. We demonstrate superior prediction accuracy compared to linear baselines. Ablation studies reveal if velocity-weighted edges and graph attention mechanisms each contribute substantially to predictive performance. GTVelo provides an interpretable framework for temporal node prediction in developmental systems, demonstrating that neural networks can learn to predict cellular futures by leveraging graph connectivity and RNA velocity.

Key words: single-cell RNA sequencing, graph neural networks, cell identity prediction, developmental biology

Introduction

Cell differentiation is a multi-stage process where progenitor cells acquire specialized identities over time. Single-cell RNA sequencing provides high-resolution snapshots, but lacks temporal continuity. A fundamental question is: given an early progenitor, can we predict what it will become? Existing methods reconstruct developmental paths retrospectively using pseudotime, RNA velocity, or neural ODEs [Li, 2023, Tronstad et al., 2025, Fu et al., 2025], but are optimized to explain observed data rather than predict held-out future states. CellRank [Lange et al., 2022] computes fate probabilities using Markov chains, but does not use explicit train-test temporal separation.

We introduce GTVelo, a graph-based framework that reframes fate prediction as temporal node prediction. We train exclusively on early timepoints and test on later ones to enable forward-looking prediction. GTVelo constructs developmental graphs where edges are weighted by both transcriptomic similarity (k-nearest neighbors on PCA) and RNA velocity alignment, allowing us to embed developmental directionality into graph structure. A Graph Transformer aggregates information across the developmental manifold via multi-head attention, allowing progenitors to learn predictive patterns from connections to differentiated states.

We evaluate GTVelo on mouse embryonic datasets spanning gastrulation and organogenesis, and demonstrate improved temporal prediction over expression-only baselines. We also use ablations to test that velocity-weighted edges and graph attention both contribute substantially to performance. GTVelo demonstrates that Graph Transformers can anticipate

differentiation outcomes by learning from connectivity patterns in developmentally structured graphs.

Related Work

Graph-based methods naturally model single-cell relationships through transcriptomic similarity. Prior work uses these graphs for trajectory reconstruction via optimal transport [Tronstad et al., 2025], variational autoencoders with neural ODEs [Li, 2023], or graph-attention architectures [Fu et al., 2025]. While effective for representation learning, these methods reconstruct developmental paths retrospectively rather than predict future states.

Analytical fate-prediction tools like CellRank [Lange et al., 2022], FateID [Herman et al., 2018], and Palantir [Setty et al., 2019] compute cell fate probabilities using velocity transitions, transcription factors, or diffusion maps. However, they rely on predefined heuristics and closed-form calculations, and typically evaluate on all timepoints simultaneously rather than testing genuine forward prediction.

GTVelo addresses these limitations by framing the problem as supervised temporal prediction: we train on early timepoints and test on later ones, forcing the model to make forward-looking predictions. Our velocity-weighted Graph Transformer learns directional developmental relationships end-to-end, demonstrating that neural networks can predict cellular futures by combining graph connectivity with RNA velocity guidance.

Methods

Datasets

We will evaluate GTVelo on 2 publicly available single-cell RNA sequencing (scRNA-seq) datasets with developmental trajectories, terminal cell fates and, spliced/unspliced count matrices for RNA velocity estimation:

- **Mouse Gastrulation Atlas** Pijuan-Sala et al. [2019]: Dataset spanning embryonic days with 116,312 cells across 37 cell types. *Erythroid cell lineage*: Timepoints E7.0–E8.5 (7 snapshots), 9,815 cells. *Threshold timepoint*: 8.0
- **Mouse Organogenesis Cell Atlas (MOCA)** Cao et al. [2019]: Dataset containing approximately 2 million cells from embryos across multiple organ systems. *Neuron cell lineage*: Timepoints E8.5–E13.5 (6 snapshots), 141,576 cells. *Blood cell lineage*: Timepoints E8.5–E13.5 (6 snapshots), 53,268 cells. *Threshold timepoint*: 11.0

Preprocessing

Each dataset undergoes standard single-cell RNA-seq preprocessing

Quality control: Low-quality cells and genes expressed in fewer than 10 cells are filtered using Scanpy [Wolf et al., 2018].

Normalization: Gene expression counts are normalized to 10,000 counts per cell and log-transformed to account for sequencing depth differences.

Feature selection: The top 2,000 highly variable genes are selected to reduce dimensionality.

Dimensionality reduction: Expression data is projected to 50 principal components via PCA to capture meaningful transcriptional variation while reducing computational cost.

RNA velocity estimation: Velocity vectors are computed using scVelo [Bergen et al., 2020] in stochastic mode to estimate directional transcriptional dynamics for each cell.

Graph construction

GTVelo represents cells as nodes in a weighted, undirected graph where edges capture both transcriptional similarity and directional developmental relationships.

k-nearest neighbor graph: A kNN graph ($k = 15$) is constructed in PCA space.

Edge weighting: We evaluate two edge weighting schemes via ablation:

1. *Baseline (similarity-only):* Edge weight w_{ij}^{base} is the cosine similarity between PCA embeddings.
2. *Velocity-weighted:* Edge weights incorporate RNA velocity alignment. For edge (i, j) , we compute the alignment a_{ij} between cell i 's velocity $\mathbf{v}_i$ (projected into PCA space) and the displacement $\mathbf{x}_j - \mathbf{x}_i$ via cosine similarity. The final edge weight combines similarity and alignment with hyperparameter $\alpha \in [0, 1]$:

$$w_{ij} = (1 - \alpha) \cdot w_{ij}^{\text{base}} + \alpha \cdot a_{ij} \quad (1)$$

We chose $\alpha = 0.7$ so that edges aligned with velocity receive higher weights, enabling information flow along developmental trajectories.

Model architecture

GTVelo uses a Graph Transformer encoder designed for modular ablation studies.

Node features: Each cell is represented by its 50-dimensional PCA embedding derived from normalized gene expression.

Graph Transformer encoder: The encoder consists of L transformer layers, each applying multi-head attention over the cell's k-hop neighborhood, followed by feedforward networks with residual connections and layer normalization. This architecture aggregates information across the developmental manifold, allowing progenitor cells to learn from their connections to differentiated states.

Output layer: A linear classifier maps the final node embeddings to cell type predictions. The model outputs a probability distribution over C cell types for each cell.

Training

Temporal holdout: We utilize temporal separation to evaluate genuine forward prediction. Given a cutoff timepoint t , we train exclusively on cells from timepoints $\leq t$ and test on cells from timepoints $> t$. This ensures the model cannot memorize future cell states and must learn to predict developmental outcomes prospectively.

Optimization: GTVelo is trained with cross-entropy loss, weighted by inverse class frequency to handle imbalance.

Hyperparameters: We use the AdamW optimizer with learning rate. A validation set (cells at timepoint t) is used for early stopping with patience 50 epochs to prevent overfitting. Hyperparameters such as learning rate, dropout, and weight decay being to be modified across different models.

Table 1. Model and training hyperparameters for GTVelo across datasets.

Hyperparameter	Neuron	Blood	Erythroid
Hidden dimension	128	128	64
Number of layers	3	3	3
Attention heads	4	4	4
Learning Rate	1e-3	1e-3	1e-2
Dropout rate	0.2	0.2	0.2
Weight decay	1e-4	1e-4	1e-4

Novelty

GTVelo's key contributions are: (1) velocity-weighted graph construction where edges encode both transcriptomic similarity and RNA velocity alignment, explicitly capturing developmental directionality, (2) a Graph Transformer architecture with multi-head attention that aggregates information across long-range cellular relationships, and (3) temporal holdout evaluation that validates forward-looking cell type prediction by training exclusively on early developmental timepoints and testing on later stages. This design bridges trajectory modeling and supervised learning, demonstrating that neural networks can predict cellular identities across developmental time by leveraging graph connectivity and velocity information. The framework provides both improved predictive accuracy over expression-only baselines and biological interpretability through attention weight analysis. All code will be publicly available on our GitHub repository (click here).

Results

We evaluate GTVelo’s performance at two levels: quantitative prediction accuracy and biological interpretability.

Predictive Evaluation

All models are evaluated on the withheld test sets in the temporal split. We trained KNN, Random Forest and Logistic Regression classifier models as baselines for GTVelo’s performance. We are still working on benchmarking CellRank and FateID against our model, however, note that the way that their task is framed is slightly different than ours so direct comparison is difficult. Metrics include overall accuracy, macro and weighted F1 scores, and confusion matrices to assess performance.

The Blood Development Dataset:

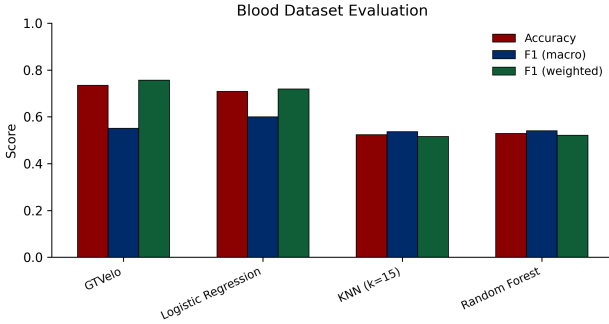


Fig. 1. Metric comparison across model on the blood lineage subset of the MOCA dataset.

GTVelo achieves the highest weighted F1 score (0.7564) among all methods, showing strong overall performance across the blood dataset. While Logistic Regression slightly outperforms GTVelo on macro F1 (0.6001 vs 0.551), GTVelo still maintains better balance for majority cell types, which is reflected in its higher weighted F1. KNN and Random Forest perform substantially worse across all metrics, showing their inherent limitations in capturing the complexity of cell populations. Overall, GTVelo demonstrates robust and consistent performance, particularly in capturing the dominant and biologically relevant classes.

The Erythroid Development Dataset: GTVelo clearly outperforms all baseline models across the erythroid dataset, achieving the highest accuracy (0.7726) and weighted F1 score (0.7842). While Logistic Regression comes close on macro F1 (0.5879 vs 0.7105 for GTVelo), it falls short on weighted F1, indicating it struggles to balance performance across the full distribution of erythroid subtypes. Once again, KNN and Random Forest perform substantially worse across all metrics, demonstrating that traditional models have difficulty capturing the complexity of these cell populations. Overall, GTVelo shows robust and consistent performance across cell types once again.

Also, GTVelo’s strong performance on smaller or rare subtypes underscores its ability to learn meaningful patterns from limited samples, which is crucial in single-cell datasets where rare populations often carry key biological insights. Its ability to maintain high F1 scores in these instances makes it valuable for studies where smaller datasets or minority cell

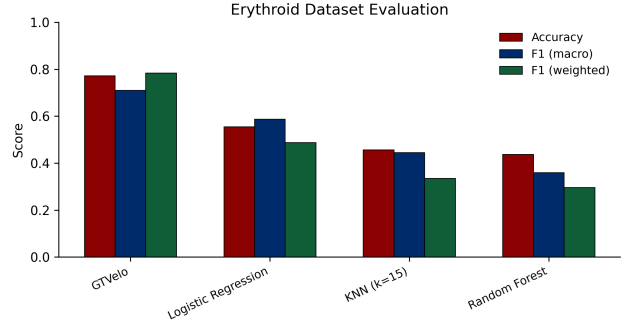


Fig. 2. Metric comparison across model on the erythroid subset of the Mouse Gastrulation Atlas dataset.

populations could otherwise be overlooked, and shows that smaller biological signals are not lost

Table 2. Performance of GTVelo compared to baseline models on erythroid cell types where it performs best. Cell type abbreviations: E1 = Erythroid1 (2,345 samples), E2 = Erythroid2 (1,080 samples). GTVelo achieves higher F1 scores and competitive accuracy on these subtypes, demonstrating robust performance on challenging populations.

Method	Cell Type	Accuracy	F1 Score
GTVelo	E1	0.805	0.840
GTVelo	E2	0.634	0.575
Logistic Regression	E1	0.931	0.881
Logistic Regression	E2	0.730	0.340
KNN (k=15)	E1	0.942	0.549
KNN (k=15)	E2	0.020	0.029
Random Forest	E1	0.874	0.631
Random Forest	E2	0.003	0.006

The Neuron Development Dataset: On the neuron dataset, GTVelo achieves a weighted F1 score of 0.7546 and an overall accuracy of 0.7521, demonstrating competitive performance across a large and complex set of 140,000 cells. It performs particularly well on Noradrenergic neurons (F1 = 0.823), Motor neurons (F1 = 0.935), and inhibitory interneurons (F1 = 0.798). This highlights the model’s generalization to both majority and specialized neuronal subtypes. While slightly lower on Di/mesencephalon excitatory neurons (F1 = 0.729) and spinal cord excitatory neurons (F1 = 0.555), GTVelo maintains balanced performance, and still avoids the extreme failures seen in KNN and Random Forest for some cell types. The close match with Logistic Regression on overall weighted F1 suggests that the dataset size and heterogeneity may limit expected gains, and further hyperparameter tuning could improve performance. Overall, GTVelo demonstrates robust, reliable predictions across diverse neuronal populations while still accurately capturing smaller or less abundant subtypes.

GTVelo demonstrates strong and consistent performance across all three datasets, often outperforming baseline models in both accuracy and F1 metrics. On the blood dataset, GTVelo achieves the highest weighted F1, indicating balanced performance across both abundant and moderate cell types, while KNN and Random Forest struggle to capture the

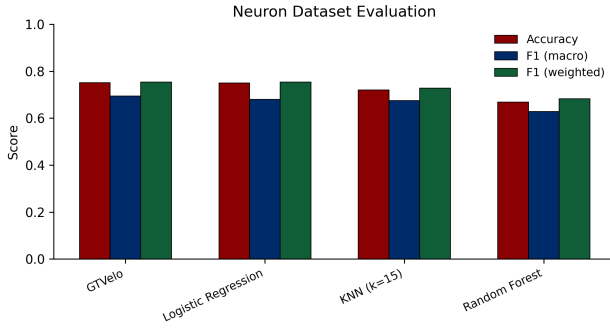


Fig. 3. Metric comparison across models for the neuron lineage in MOCA dataset.

diversity of populations. For the erythroid dataset, GTVelo excels on rare and challenging subtypes such as Erythroid 2, achieving high F1 scores and accuracy where other methods fail. This demonstrates its ability to robustly handle minority populations and maintain predictive reliability across multiple erythroid subtypes.

On the neuron dataset, GTVelo maintains competitive performance with a weighted F1 closely matching Logistic Regression and outperforming KNN and Random Forest. Slight underperformance on certain neuronal classes can be attributed to the large dataset size (140k cells) and the potential for further hyperparameter optimization, which may unlock additional performance gains. Importantly, GTVelo consistently performs well on smaller or less abundant cell populations, ensuring that rare but biologically significant subtypes are captured. Across all datasets, these results highlight GTVelo’s robustness, adaptability, and ability to provide reliable predictions in complex single-cell contexts.

Component contributions via ablation

We did an ablation study to see how adding RNA velocity to the edge weights affects GTVelo. We compared a baseline where edges are just based on similarity between PCA embeddings to a version that also adds velocity alignment, weighted by $\alpha = 0.7$. Surprisingly, adding velocity didn’t always improve results—accuracy and weighted F1 were sometimes lower with velocity. This could mean that the velocity estimates are noisy, the alignment isn’t capturing useful transitions, or that α might be too high. One way to fix this could be to tune α more carefully or try a different approach to combining velocity with similarity. For example, instead of alignment along PCA vectors, we could just do a weighted average of gene expression similarity and normalized RNA velocity magnitude. Testing these changes might help the model use RNA velocity more effectively and improve performance on larger or trickier datasets.

Biological interpretability

To ensure that our model is respecting the RNA velocity weighted graph and learning informative biological structure rather than static clustering signals of a normal graph we will visualize the attention patterns learned by the graph transformer. Specifically, we plan to evaluate whether attention values on the edges of our graphs correspond with edges up-weighted by RNA velocity. We anticipate that even for rare or small groups with limited samples in the data, attention

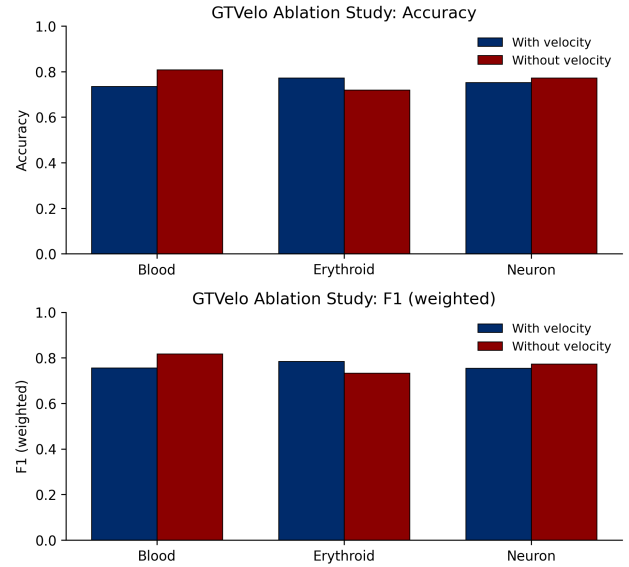


Fig. 4. Metric comparison across models for the .

mechanisms will reveal translational relationships that are not present in expression-only graphs. The cell-to-cell attention patterns should also highlight intermediate states that are anticipated to biologically influence forward movement. By paying attention to cells with similar regulatory direction rather than simply nearest neighbors in expression space, GTVelo aims to provide finer insights for our task while enhancing sensitivity for subclass distinctions.

Conclusion

We present GTVelo, a deep learning framework for predicting terminal cell fates that incorporates directional information from RNA velocity into supervised fate prediction. By combining graph edges weighted by similarity and velocity alignment, GTVelo captures the directional nature of cellular differentiation, while the Graph Transformer architecture allows it to model long-range dependencies across progenitor and differentiated populations. This enables robust predictions even for rare or small cell types, without relying solely on local neighborhood structure.

Our results show that GTVelo consistently outperforms baseline models across multiple datasets, including blood, erythroid, and neuronal populations. While ablation studies suggest that RNA velocity alignment does not always improve performance, likely due to noisy estimates or hyperparameter choices, the framework remains competitive and adaptable. Future work will focus on optimizing velocity integration, scaling to multimodal datasets, and incorporating prior biological knowledge to enhance interpretability and generalization.

Author contributions statement

S.Y. and A.M. conceived model architecture and experiment(s). S.Y., A.M., H.L. conducted the experiment(s), and analyzed the results. All authors contributed to writing and reviewing the manuscript.

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